



Wisconsin School of Business

General Business 730: Prescriptive Modeling & Optimization for Business Analytics
Fall 2021
2-credits

Instructor:	Jordan Tong
Office:	4293 Grainger
Office Hours:	Typically Mondays 3:15-4:45PM. See weekly pages on website for current info.
Email:	jordan.tong@wisc.edu
Teaching Assistant:	Univa Song
Email:	univa.song@wisc.edu
Office Hours:	Typically Fridays 10:30 – Noon; See weekly pages on website for current info.
Course Title:	Prescriptive Modeling & Optimization for Business Analytics – General Business 730
Class Meetings:	Please refer to the course website for up-to-date information. This course releases new material twice a week: • Mondays: Asynchronous video lecture (~2 hours recommended; I will post them and you can complete them on your own time) • Thursdays (Section 1: 11AM-1:30PM in Grainger 2335; Section 2: 2:30-5PM in Grainger 2520) Face-to-face class meetings. I will also attempt to record these face-to-face meetings.
Credit Hours:	This is a two credit class. These credits are met according to the traditional Carnegie Definition: This class meets twice a week for 3-4 hours of activities and carries the expectation that students will work on additional course learning activities (reading, writing, problem sets, studying, etc) for about 6 hours per week. The syllabus includes more information about meeting times and expectations for student work. See the UW-Madison Credit Hour Policy .
Canvas:	https://canvas.wisc.edu/courses/272471
Course Description:	This course will introduce you to fundamentals of prescriptive business analytics. Specifically, you will gain basic proficiency in modeling and optimization using spreadsheets (in Excel) and programming languages (in Python). The course is designed primarily in a learning-by-doing format. You will learn some fundamentals of optimization theory, but the course is largely problem-driven and prioritizes application and problem-solving savvy. The course is only 2-credits, so you will not leave this class a seasoned expert. Rather, my goal is for you to develop an understanding of the kinds of opportunities modeling or optimization provides. I hope to also give you enough practical experience to be able recognize problems you can work on productively yourself and to converse intelligently with seasoned experts you may recruit for help.

This Version: 10/2/2021, see course webpage for most up-to-date information.

Although the course is primarily problem-driven, it is still divided into modules. The first module focuses on helping you develop practical skills in spreadsheet modeling, which is ubiquitous in business analytics, as well as some general capabilities for building models in Python. The second module focuses on linear optimization, solved using spreadsheets and Python. The third module focuses on problems that require integer optimization. Lastly, time permitting, we will discuss some other prescriptive analytics current topics or tools.

Requisites: Graduate standing; Enrolled in the Masters of Science Business Analytics or permission from the instructor.

Requisites Comments: While the course is designed for MSBA students who have also taken the “statistics and coding bootcamp,” other non-MSBA students are welcome. I generally assume students have the following base knowledge, which are therefore recommended but not formally required:

- General familiarity with computers (mac or PC is ok).
- Basic proficiency with Excel.
- Introductory-level experience coding in Python or similar coding language.
- Basic knowledge of probability, statistics, and some linear algebra.

Learning Outcomes

When you successfully complete this course, you will be able to:

1. Identify appropriate modeling, analysis, and optimization methodologies when faced with a concrete problem.
2. Leverage spreadsheet software (e.g., Excel) and programming languages (e.g., Python) for modeling and optimization.
3. Classify optimization problems (e.g., linear, integer, nonlinear, ...etc.) and explain the challenges of solving different problem types.
4. Recognize classic optimization problems (e.g., knapsack, network, traveling salesman,...etc) and recall how to formulate them.
5. Discuss basic optimization theoretical concepts and their practical implications.

Required Materials

- Laptop computer (Mac or Windows OK)
 - Microsoft Excel 2019 or later.
 - Python -- Anaconda distribution, with Jupyter notebooks. Installation instructions will be distributed in class.

You must have your own working laptop computer throughout the entirety of the semester to succeed in this course. Your computer must be able to run reasonably-sized Excel and Python optimization programs effectively. I cannot provide computer support. If you run into a major computer issue, here are some links that may be helpful regarding computer rentals: DoIT Help Desk Rentals: <https://it.wisc.edu/doit-computer-lending-program-request/> Library Rentals: <https://www.library.wisc.edu/college/services-at-college/equipment/laptops-and-tablets/> I also understand that WSB lab computers have Python Anaconda installed.

- You are not required to purchase any books or course packets for this course. All necessary material will be accessible either in class or via the course website.

Grading & Assessment

Category	Weight
Participation and Participation Assignments	20%
Graded Individual Assignments	60%
Modeling Fundamentals	20%
Linear Optimization	20%
Integer Optimization	20%
Application Project	20%

The above percentages will be used to compute a total weighted average score for each student in the class. These total scores will then be ranked, and potential grades will be computed according to two different methods. A student's final grade will be the higher of the grades obtained by these methods.

Method 1: The first method uses the raw total as follows:

Over 93%: A
86%-92.99%: AB
78%-85.99%: B
70%-77.99%: BC
60%-69.99% C
50%-59.99% D
Below 50% F

Method 2: The second method uses the percentile performance within the class as follows:

Top 30% of class: A
Next 30% of class: AB
Next 30% of class: B
Bottom 10% of class: Grade based solely on first method

Participation and Participation Assignments

Please bring your laptop to every F2F meeting. Active participation for each session is expected, although note that there is no explicit grade for attendance. Your participation grade will be primarily determined by the *participation assignments* that are conducted in and between classes. Active and *helpful participation in class or the discussion boards* may also affect your grade in borderline cases. While attendance is not directly required or incentivized, it is encouraged and we often spend time working in break-out groups on that associated participation assignment.

Participation assignments will accompany most class meetings. They are your opportunity to practice and collaborate while being graded primarily on effort and completeness, not correctness. You are encouraged to consult your classmates on these assignments, but you cannot simply "copy" files or portions of files (i.e., your assignments should not be identical). Participation assignments are graded on a 0, 1, 2 basis:

Points	Interpretation
2	Makes it obvious that the student is following along in class. Though mistakes are allowed, documents clear and sincere effort on every question. Everyone can achieve and expect this feedback with sincere effort.
1	Incomplete/noticeably below peers in documented effort OR some effort, but there is concern that the student is not following along in class

0	No submitted files or documented effort (i.e., clicking things randomly or typing in random numbers is not sufficient to get partial credit).
---	---

I will calculate your participation points (out of 2 points per participation assignment) and drop the lowest score.

If at some point during the term you find that you cannot attend a particular session, you are responsible for obtaining any handouts or notes, submitting any assignments, and mastering the material for that session. All material is available online, and I will attempt to record in-person lectures – but I still recommend checking in with a classmate.

Individual Assignments

Individual assignments should be completed on your own and your work should be your own. To be fair to all students, you are **only** permitted to ask or receive help regarding individual assignments through the course discussion boards, or with me in office hours. Giving or receiving help outside the discussion board is considered academic dishonesty (see policy below). Giving or receiving help is considered academic dishonesty and will result in failure in the course (see “Academic Dishonesty” below).

Application Project

An application project is due at the end of the class (see course website for exact dates). You may complete this project by yourself or with one other student enrolled in the class. Details will be distributed in class. It should be your own (or you and your teammate’s own) work. While you may consult others for feedback and advice, you should be able to claim the project as your own intellectual product. You must also properly cite all references (e.g., copying a project you found online or elsewhere without proper citation is considered academic dishonesty). Such academic dishonesty will result in failure in the course (see “Academic Dishonesty” below) and advancement for further disciplinary action from the School of Business or the University of Wisconsin-Madison.

Late Work and Other Special Circumstances

Assignments submitted online after the due date will be automatically penalized from the total possible points as follows:

10% if submitted less than 24 hours late.

25% if submitted between 24 hours and 1 week late.

50% if submitted more than 1 week late but before the last day of class.

No late work may be submitted after the last day of class. Please contact me if there are extreme circumstances. Computer issues or technical difficulties are not an extreme circumstance.

Academic Dishonesty

Academic integrity is an important element of this and all classes at the University of Wisconsin-Madison. For the individual assignments, all work is to be done on an **individual** basis. You may not discuss or exchange information regarding those assignments with others either inside or outside of this class except via the discussion board on the course website. Similarly, for the individual application project, all work should be your own.

Academic dishonesty is grounds for failure in the course. Leveraging material from previous years is also considered academic dishonesty. All incidents of academic dishonesty may result in failure in the course and may be referred to the Dean of Students for disciplinary action. More information can be found online at

<http://www.students.wisc.edu/doso/students/>. If there is any question as to whether an activity is or is not permissible (in this class), consult me prior to undertaking the activity.

By enrolling in this course, each student assumes the responsibilities of an active participant in UW-Madison’s community of scholars in which everyone’s academic work and behavior are held to the highest academic integrity standards. Academic misconduct compromises the integrity of the university. Cheating, fabrication, plagiarism, unauthorized collaboration, and helping others commit these acts are examples of academic misconduct, which can result in disciplinary action. This includes but is not limited to failure on the assignment/course, disciplinary probation, or suspension. Substantial or repeated cases of misconduct will be forwarded to the Office of Student Conduct & Community Standards for additional review. For more information, refer to studentconduct.wiscweb.wisc.edu/academic-integrity/.

This Version: 10/2/2021, see course webpage for most up-to-date information.

COVID-19 University of Wisconsin-Madison Policies

Please refer to <https://covidresponse.wisc.edu/what-to-expect-this-fall/> for up-to-date university policies.

Also, please note the following face-covering policy from campus: “We expect everyone on campus – students, faculty, staff, and visitors – to follow the reasonable and research-based measures we have in place to protect our community. Those who can wear a face covering but refuse to do so will be asked to leave the building. Any absence from class due to refusal to wear a mask will be treated by the instructor as an unexcused absence, with consequences as indicated on individual class syllabi. Students who repeatedly fail to comply will be referred to the Office of Student Conduct and Community Standards and may be subject to disciplinary action under [the non-academic misconduct policy](#). Students who have a medical condition or disability that affects their ability to wear a face-covering should [obtain an accommodation](#).”

Information about Recordings

I plan to record lectures, when possible. Please refer to <https://registrar.wisc.edu/ferpa-facstaff/> for information about privacy related to these recordings. In general, please also note that you are not allowed to copy or record lecture materials and post them outside of class. See <https://registrar.wisc.edu/ferpa-facstaff/> for details.

Course Evaluations

Students will be provided with an opportunity to evaluate this course and your learning experience. Student participation is an integral component of this course, and your feedback is important to me. I strongly encourage you to participate in the course evaluation. UW-Madison now uses an online course evaluation survey tool, AEFIS. In most instances, you will receive an official email two weeks prior to the end of the semester when your course evaluation is available. You will receive a link to log into the course evaluation with your NetID where you can complete the evaluation and submit it, anonymously. Your participation is an integral component of this course, and your feedback is important to me. I strongly encourage you to participate in the course evaluation.

Academic Calendar and Religious Observances

See: <https://secfac.wisc.edu/academic-calendar/#religious-observances>.

Accommodations for students with disabilities:

The University of Wisconsin-Madison supports the right of all enrolled students to a full and equal educational opportunity. The Americans with Disabilities Act (ADA), Wisconsin State Statute (36.12), and UW-Madison policy (Faculty Document 1071) require that students with disabilities be reasonably accommodated in instruction and campus life. Reasonable accommodations for students with disabilities is a shared faculty and student responsibility.

Students are expected to inform me of their need for instructional accommodations by the end of the third week of the semester, or as soon as possible after a disability has been incurred or recognized. I will work either directly with you or in coordination with the McBurney Center to identify and provide reasonable instructional accommodations. Disability information, including instructional accommodations as part of a student's educational record, is confidential and protected under FERPA.

Wisconsin School of Business Commitment to Diversity:

Diversity and inclusion are core values of the Wisconsin School of Business. We believe in the importance of learning from others whose experiences and perspectives differ from our own. Students, faculty and staff are expected to embody inclusive behaviors that create an exemplary learning community for all.

If you would like to learn more, or have suggestions for how we can improve the climate for learning, please contact your instructor or TA directly or the Director of Diversity and Inclusion, Binu Hill at binu.palta.hill@wisc.edu, 263-3462, 4250 Grainger Hall.

This Version: 10/2/2021, see course webpage for most up-to-date information.

Week #	Date of Synchronous Meeting	Module	Approximate assignments in progress; refer to course website for exact due dates and times.
1	10/14	Modeling Fundamentals	
2	10/21	Modeling Fundamentals	Modeling Fundamentals
3	10/28	Modeling Fundamentals	
4	11/4	Linear Optimization	Linear Optimization
5	11/11	Linear Optimization	
6	11/18	Integer Optimization	Integer Optimization
7	11/25	Thanksgiving	
8	12/2	Integer Optimization	
9	12/9	Other Topics/Application Project	Application Project
10	No Meeting (Office Hours only)	Other Topics/Application Project	